

Lab 2 : MultiCycle Performance

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0.1 Matrix Multiplication Performance

0.1.1 Procedure

1. Compile the matrix multiplication program in the OTTER-memory mapped version of RARS. The machine code file needs to be saved.
2. Load the mem.txt file into the multicycle OTTER in Vivado and simulate.
3. Capture the waveforms for the matrix multiplication to prove the correct execution.
4. Provide CPU execution time , Power, and Area.

0.2 Testall Performance

0.2.1 Procedure

1. Run the testall program and capture the waveforms and metrics from the MM performance

0.3 Notes

0.3.1 RARS

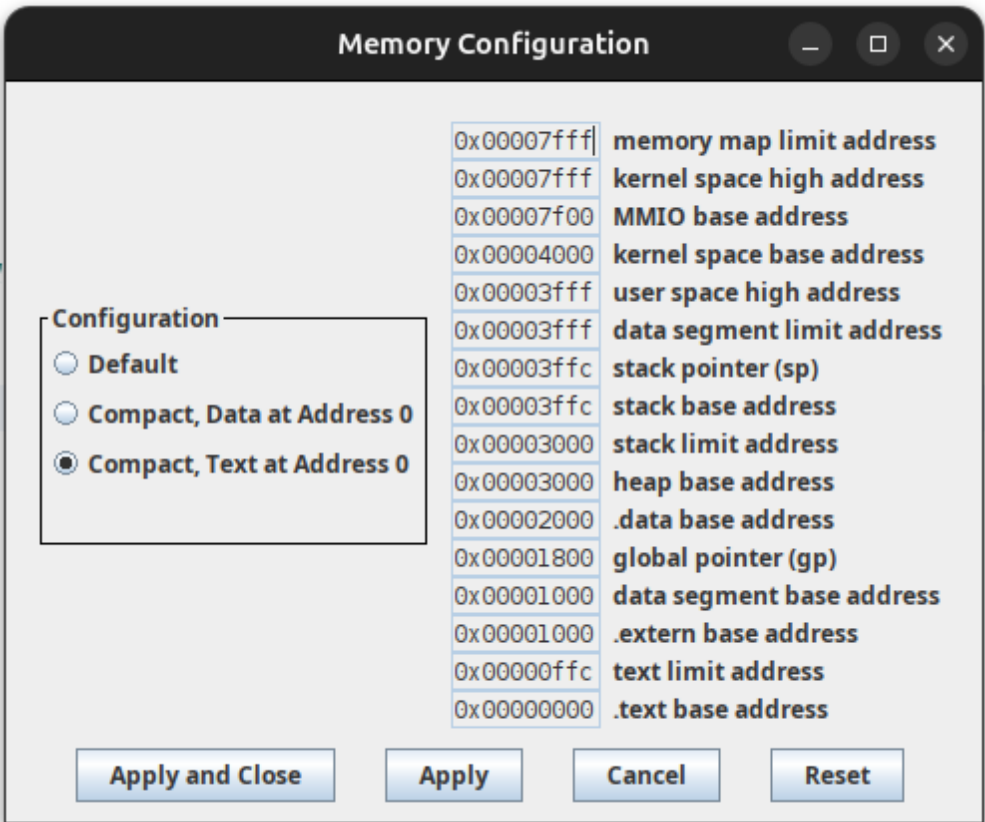


Figure 1: RARS Memory Config Settings

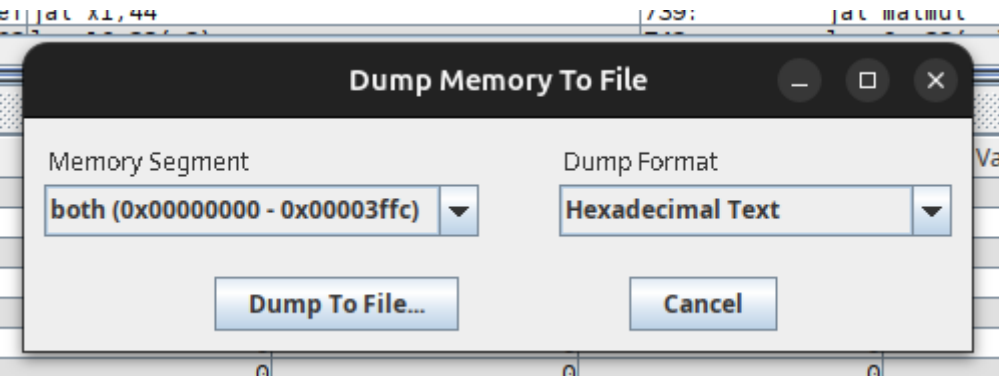


Figure 2: RARS Memory Dump Config